

AFT Models & Best Subset Selection via MIP

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Agenda

1> AFT Model Overview

2> AFT Literature Review

Huang, Ma & Xie (2006)

Wang & Wang (2010)

Khan & Akhter (2024)

3> BSS

4> BSS via MIP

Bertsimas and King- Best Subset Selection via a
Modern Optimization Lens

Hazimeh, H., & Mazumder, R. (2020).

Bertsimas, D., Dunn, J., Gibson, E., & Orfanoudaki, A. (2022)

1> AFT Model Overview

What is AFT Model? $\log(T) = X\beta + \varepsilon$

ε follows a parametric distribution (Weibull, log-normal, log-logistic, etc.)

<Idea>

- No PH assumption required (Unlike Cox PH)
- Parametric survival model; log Survival time as a linear function of covariates
- Covariates directly accelerate or decelerate time to event

<Interpretation> (TR) = $\exp(\text{Beta})$

- $\text{TR} > 1$: Time to event increases (acceleration effect)
- $\text{TR} < 1$: Time to event decreases (deceleration effect)

- Variable selection is critical in high-dimensional settings; $p \gg n$
- Challenge: **Censoring** must be handled with variable selection → solved via KM weights (Stute) / iterative imputation (Buckley–James)

2> AFT Literature Review

- ① Huang, Ma & Xie (2006):
Abstract: Lasso & TGDR for High-Dimensional AFT

Semiparametric estimation in the AFT model with an unspecified error distribution

1> Buckley-James estimator : adjusts censored observations using the Kaplan- Meier estimator.

2> Rank based estimator: motivated from the score function of the partial likelihood

2> AFT Literature Review

- 1 Huang, Ma & Xie (2006):
Abstract: Lasso & TGDR for High-Dimensional AFT

Background & Motivation

- AFT model: log failure time as linear function of covariates — intuitive alternative to Cox PH
- Microarray studies: $n \sim$ hundreds, $p \sim$ thousands — identifying survival-associated genes is fundamentally a variable selection problem that existing methods cannot handle

Reference: Huang, Jian, Ma, Shuangge and Xie, Huiliang. (2006). Regularized estimation in the accelerated failure time model with high-dimensional covariates. *Biometrics* 62(3), 813–820.

2> AFT Literature Review

- **Method: Two regularization approaches** within Stute's weighted least squares framework

Lasso — convex, computationally efficient gradient-directed algorithm

TGDR (Threshold Gradient Directed Regularization) — flexible non-convex shrinkage

Censoring handled via [Kaplan–Meier weights](#) (not Buckley–James)

<WLS estimation in AFT Model>

$$T_i = \beta_0 + X_i' \beta + \varepsilon_i, \quad i = 1, \dots, n,$$

T_i : logarithm of the failure time $\delta_i = 1_{\{T_i \leq C_i\}}$: censoring indicator

X_i : d -dimensional covariate vector for the i th subject in a random sample of size n .

C_i : logarithm of the censoring time

If **Right-Censored**

only observe (Y_i, δ_i, X_i) with $Y_i = \min\{T_i, C_i\}$,

Weighted least squares estimator defined as $\hat{\theta} \equiv (\hat{\beta}_0, \hat{\beta})$ that minimizes

$$M(\theta) = \frac{1}{2} \sum_{i=1}^n w_{ni} (Y_{(i)} - \beta_0 - X'_{(i)} \beta)^2.$$

2> AFT Literature Review

$$w_{n1} = \frac{\delta_{(1)}}{n}, \text{ and } w_{ni} = \frac{\delta_{(i)}}{n - i + 1} \prod_{j=1}^{i-1} \left(\frac{n - j}{n - j + 1} \right)^{\delta_{(j)}}, \quad i = 2, \dots, n, \quad \text{Incorporate Kaplan-Meier weights}$$

Stute estimator transforms the censored survival problem into a standard least squares framework

Why Stute over Classical censored AFT?

computationally infeasible even at moderate $p \rightarrow$ Cannot scale to high dimensions



1. Rigorously theoretical justifications under reasonable assumptions

(Consistency, Asymptotic normality)

2. Due to least squares structure in the criterion function for the Stute

estimator, **computationally efficient** to apply in AFT rather than Cox

2> AFT Literature Review

Regularized weighted LS regression for AFT model

Using the weighted centered values, the intercept estimate is zero.

$$M(\beta) = \frac{1}{2} \sum_{i=1}^n (Y_{(i)} - X'_{(i)}\beta)^2.$$

1> Lasso

- Minimize weighted LS objective under L_1 constraint: $\sum_{j=1}^d |\beta_j| \leq u$,
- Implemented via L1 boosting algorithm
(Kim & Kim, 2004)— convergence rate independent of dimension $\rightarrow$ valuable for high-dim genomic data
- Tuning parameter u selected by AIC score

2> TGDR

- At each iteration, only updates coefficients whose gradient exceeds $\tau \times \max$ gradient
- τ controls sparsity: large $\tau \rightarrow$ LASSO-like (sparse) / small $\tau \rightarrow$ Ridge-like (dense)
- Capable of selecting correlated covariates simultaneously (LASSO cannot) Tuning: k by CV score, τ by AIC score (two-step) $0 \leq \tau \leq 1$: fixed threshold

2> AFT Literature Review

- **Result:** Simultaneous variable selection and estimation feasible under high-dimensional censored AFT
- **Limitation:** No selection consistency guarantees for Lasso; TGDR non-convex → heavier computation

2> AFT Literature Review

- ② Wang & Wang (2010)
Abstract: Boosting-Based AFT Estimation

[Advantages of AFT]

1. AFT model has a linear regression form, typically based on a logarithmic transformed response variable
2. Proportional assumption may not be realistic in some settings
3. BJ estimation for the AFT model can be conveniently extended to describe more complex data structures

Reference: Wang, Zhu and Wang, C.Y. (2010). Buckley-James boosting for survival analysis with high-dimensional biomarker data. *Statistical Applications in Genetics and Molecular Biology* 9(1), Article 24.

2> AFT Literature Review

Huang et al. (2006)

(LASSO) (Tibshirani, 1996) (TGDR) (Friedman and Popescu, 2004),

→ (IPC) weighted least squares estimator (Stute, 1993).
different from BJ iterative imputation, and can produce
inferior prediction accuracy

Datta et al. (2007)

partial least squares (PLS) and LASSO by mean imputation

Wang et al. (2008a)

elastic net algorithm (Zou and Hastie, 2005)

(Huang and Harrington, 2005),

PLS with BJ PLS can't conduct variable selection.

Developed for describing **simple linear effects**

→ Genetic architecture of the common diseases is very complex

& diseases are produced by the **nonlinear** interaction of genetic and environmental covariates

2> AFT Literature Review

“Unified framework on regression models for a time-to-event outcome with the gene expression data”

This paper is motivated to develop a unified framework to model survival times with gene expression microarrays. The proposed approach can fit linear effects model, nonlinear effects model, and nonlinear effects model with high order nonlinear interactions. A key component of this framework relies on boosting tech-

- **Motivation:** Buckley–James estimator is computationally burdensome → limits scalability to high dimensions
- **Method:** Incorporate boosting into the Buckley–James framework
 - Iteratively imputes censored observations at each boosting step
 - Variable selection performed simultaneously via boosting's implicit shrinkage (early stopping)

2> AFT Literature Review

- **Key contribution:** First to make high-dimensional AFT feasible under the Buckley–James framework

BJ iterative estimation method(BJ + Boosting)

BJ estimator is closely related to the ordinary least squares estimator without censoring

Boosting is a method for minimizing convex loss functions via gradient descent techniques

Captures nonlinear effects and complex interactions via the combination of the Buckley-James estimator and boosting

- **Limitation:** Sparsity induced implicitly via early stopping — approximate, not exact variable selection; selection inconsistency possible

2> AFT Literature Review

③ Khan & Akhter (2024): Abstract: Ranking-Based Variable Selection for Censored AFT

Motivation

- Existing AFT methods (Lasso, TGDR, boosting) induce sparsity via continuous penalties → approximate shrinkage, not exact subset identification
- RBVS (Baranowski et al. 2020) achieves discrete subset selection for complete data — extend it to censored survival data via AFT

Reference: Khan, Md Hasinur Rahaman and Akhter, Marzan. (2024). Ranking based variable selection for censored data using AFT models. *Communications in Statistics - Simulation and Computation* 53(6), 2917–2939.

2> AFT Literature Review

Base Model: Stute's Weighted LS for AFT

$$l(\beta_0, \boldsymbol{\beta}) = \frac{1}{2} \sum_{i=1}^n \pi_{ni} \left(Y_{(i)} - \beta_0 - \mathbf{X}'_{(i)} \boldsymbol{\beta} \right)^2,$$

π_{ni} : jumps in the Kaplan–Meier estimator

$\hat{w}_j = \hat{w}_j(\mathbf{Z}_1, \dots, \mathbf{Z}_n), j = 1, \dots, p$: measure for giving the estimates under different techniques including penalization

Ranking measure $\hat{w}_j = |\hat{\beta}_j|$ is obtained by fitting Lasso or MC+ on the Stute weighted objective , serving as each variable's impact score for subsequent ranking

2> AFT Literature Review

Lasso estimator

$$\longrightarrow \hat{\beta}_{pen} = \operatorname{argmin}_{\beta} \left(\frac{1}{n} \sum_{i=1}^n \left(Y_i - \sum_{j=1}^p \beta_j X_{ij} \right)^2 + \sum_{j=1}^p \operatorname{pen}(|\beta_j|) \right), \quad \operatorname{pen}(M) = \lambda M$$

MC + estimator

$$\longrightarrow \hat{\beta}_{pen} = \operatorname{argmin}_{\beta} \left(\frac{1}{n} \sum_{i=1}^n \left(Y_i - \sum_{j=1}^p \beta_j X_{ij} \right)^2 + \sum_{j=1}^p \operatorname{pen}(|\beta_j|) \right)$$

$$\operatorname{pen}(M) = \lambda \int_0^M \max\{0, (1 - x/(\gamma\lambda))\} dx.$$

2> AFT Literature Review

RBVS Algorithm

- **Step 1: Subsampling.** For each $b = 1, \dots, B$, draw uniformly without replacement m -element subsets $I_{b1}, I_{b2}, \dots, I_{br} \subset \{1, \dots, n\}$ for $r = \lceil \frac{n}{m} \rceil$ times.
- **Step 2: Ranking evaluation.** Evaluate $\hat{w}_j(\mathbf{Z}_i \in I_{bl}), j = 1, \dots, p$ for each $b = 1, \dots, B$ and $l = 1, \dots, r$ and sort $\hat{w}_j(\mathbf{Z}_i \in I_{bl})$ in non-increasing order. Then record corresponding variable ranking $\mathbf{R}_n(\mathbf{Z}_i \in I_{bl})$.
- **Step 3: Estimation of k -top-ranked sets.** For each $k = 1, \dots, k_{max}$, find the k -element set $\hat{A}_{k,m}$ which is the most frequently occurring in the top of the rankings $\mathbf{R}_n(\mathbf{Z}_i \in I_{bl}), b = 1, \dots, B, l = 1, \dots, r$. Then write down the probabilities $\hat{\pi}_{m,n}(\hat{A}_{k,m}) = \frac{1}{Br} \sum_{b=1}^B \sum_{j=1}^r I(\hat{A}_{k,m} = \mathbf{R}_{n,1:k}(\mathbf{Z}_i, i \in I_{bj}))$.
- **Step 4: Estimation of the top-ranked set.** Find $\hat{s} = \underset{k=0, \dots, k_{max}-1}{\operatorname{argmin}} \frac{\hat{\pi}_{m,n}(\hat{A}_{k+1,m})}{\hat{\pi}_{m,n}(\hat{A}_{k,m})}$ as estimate of the size of the top-ranked set. Return $\hat{S} = \hat{A}_{\hat{s},m}$ which is the final estimate of S .

Important variables rank consistently at top across subsamples; irrelevant do not

IRBVS Algorithm

: an iterative extension of RBVS

- Set $\hat{S} = \emptyset$.
- Repeat
 1. Define $\mathbf{Z}_i^* = (Y_i^*, X_{ij}^*, j \in \{1, \dots, p\} \setminus \hat{S}), i = 1, \dots, n$, where Y_i^*, X_{ij}^* are the residuals left after projecting Y, X_j onto the space spanned by the covariates whose indices are in $\hat{S}$.
 2. Find $\hat{S}^*$ and $\hat{s}$, which are the output of the RBVS algorithm.
 3. Set $\hat{S} = \hat{S} \cup \hat{S}^*$.

Projects out selected variables, reruns RBVS on residuals to catch jointly-related but marginally-uncorrelated variables

2> AFT Literature Review

	Huang et al. 2006	Wang & Wang 2010	Khan & Akhter 2024
Censoring	Stute (KM weights)	Buckley–James	Stute (KM weights)
Selection Method	Lasso / TGDR	Boosting	RBVS (ranking-based)
Selection Type	Continuous (approx.)	Continuous (approx.)	Discrete (exact subset)
High-dim Feasible	✓	✓ (via boosting)	✓
Network Structure	✗	✗	✗

Appendix

Regularized weighted LS regression for AFT model

1. Initialization $\beta = (0, \dots, 0)'$.
2. With the current estimate of β , compute $\mathbf{g}(\beta)$, the negative derivative of $M(\beta)$ with respect to β . Denote the j^{th} component of $\mathbf{g}(\beta)$ as $g_j(\beta)$, $j = 1, \dots, d$.
3. Find j^* that minimizes $\min(g_j(\beta), -g_j(\beta))$. If $g_{j^*}(\beta) = 0$, then stop the iteration.
4. Otherwise denote $\gamma = -\text{sign}(g_{j^*}(\beta))$. Find $\hat{\kappa} \in [0, 1]$ that minimizes $M((1 - \kappa)\beta + \kappa \times u \times \gamma \eta_{j^*})$, where η_{j^*} is a length d vector that has the j^* th element equals to 1 and the rest components equal to 0.
5. For the j^{th} component of β : $\beta_{(j)} = (1 - \hat{\kappa})\beta_{(j)}$ for $j \neq j^*$ and $\beta_{(j^*)} = (1 - \hat{\kappa})\beta_{(j^*)} + \gamma u \hat{\kappa}$.
Let $m = m + 1$.
6. Repeat steps 2–5 until convergence or a fixed number of iterations N has been reached.

β at convergence is the LASSO estimate

1. Initialize $\beta(0) = (0, \dots, 0)'$ and $\nu_0 = 0$.
2. For the current estimate $\beta = \beta(\nu)$, compute the negative gradient $\mathbf{g}(\nu) = -\partial M(\beta)/\partial \beta$.
Denote the j^{th} component of $\mathbf{g}(\nu)$ as $g_j(\nu)$. If $\max_j |g_j(\nu)| = 0$, stop the iteration.
3. Compute the vector $\mathbf{f}(\nu)$ of length d , where the j^{th} component of $\mathbf{f}(\nu)$: $f_j(\nu) = I\{|g_j(\nu)| \geq \tau \cdot \max_j |g_j(\nu)|\}$.
4. Update $\beta(\nu + \Delta_\nu) = \beta(\nu) + \Delta_\nu \mathbf{g}(\nu) \mathbf{f}(\nu)$ and $\nu = \nu + \Delta_\nu$.
5. Steps 2–4 are repeated S times. S is taken to be a large number to yield a full parameter path.

3> Why BSS for Survival Data? (Best Subset Selection)

Standard BSS minimizes RSS (Requires knowing exact outcome y_i for all observations)

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

However

In Survival data → Problem: Censored

Right Censoring $T_i > C_i$

- If a patient drops out or the study ends, we do not observe T_i exactly
- Naively treating C_i as the event time → systematic underestimation of survival time

Solution

- Use a likelihood-based criterion that accounts for censored observations
- Maximize the probability of observing the data including censoring information
- Model selection via AIC / BIC

3> Why BSS for Survival Data?

BSS in Survival Analysis: Goal & Two-Step Process

Goal: Find the optimal subset of k variables out of p covariates that maximizes interpretability and predictive performance



Interpretability

Prediction & Overfitting

Two-Step Process:

Step 1 — Search: Enumerate all 2^p candidate subset models from p covariates

Step 2 — Evaluate & Select: Choose the single best model

among 2^p candidates using a selection criterion **objective function:** $\min_{\beta} \sum_i \ell(\beta; \text{data})$ **subject to** $\|\beta\|_0 \leq K$

4> BSS via MIP

Why MIP for BSS?

Challenge: Number of candidate subsets grows exponentially: $\binom{p}{k}$ becomes intractable as p increases

- Furnival & Wilson (1974) leaps-and-bounds algorithm practical limit at $k \approx 20$
- Led to continuous penalty methods (Lasso, Elastic Net) as practical alternatives

<https://www.stat.cmu.edu/technometrics/90-00/vol-42-01/v4201069.pdf>

<Limitations of continuous penalties>

1. Provide only approximate sparsity — not exact subset selection
2. Coefficient shrinkage bias — nonzero coefficients are pulled toward zero

4> BSS via MIP

MIP Solution

- Fast discrete algorithm $\rightarrow$ warm start for MIP solver $\rightarrow$ global optimum
- Sparser & more predictive than Lasso
- Feasible at $p \sim$ thousands, within minutes
- L0-constrained regression is now tractable

4> BSS via MIP

1

Bertsimas and King- Best Subset Selection via a Modern Optimization Lens

$$\min_{\beta} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|_q^q \quad \text{s.t.} \quad \|\beta\|_0 \leq k, \mathbf{A}\beta \leq \mathbf{b},$$

- MIO + discrete first-order optimization $\rightarrow$ optimal k-subset selection
- Provable optimality guarantees
- Outperforms Lasso in accuracy & sparsity

Reference: Bertsimas, D., King, A., & Mazumder, R. (2016). Best subset selection via a modern optimization lens. *Annals of Statistics*, 44(2), 813–852.

4> BSS via MIP

② Hazimeh, H., & Mazumder, R. (2020)

Core Motivation

- Bertsimas 2016 (MIP-BSS): minutes at $p \sim 1,000$

- Lasso (glmnet): seconds at $p \sim 10^6$

-> Close this computational gap

MIO is too slow!

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda_0 \|\beta\|_0 + \lambda_q \|\beta\|_q^q, \\ q \in \{1, 2\}$$

2 complementary algorithms:

CD (Algorithm 1): cyclic coordinate descent → converges to CW minimum

Local Combinatorial Search (Algorithm 2): swap variables in/out of active set → PSI(k)/FSI(k) minimum

Reference: Hazimeh, H., & Mazumder, R. (2020). Fast best subset selection: Coordinate descent and local combinatorial optimization algorithms. *Operations Research*, 68(5), 1517–1537.

4> BSS via MIP

3 Bertsimas, D., Dunn, J., Gibson, E., & Orfanoudaki, A. (2022)

Motivation

- Cox proportional hazards model fails to capture nonlinear effects and covariate interactions
- Existing survival tree algorithms use **greedy** splitting
- Propose MIO (Mixed-Integer Optimization) to optimize the entire tree in a single step

objective: $\min_T \text{error}(T, D) + \alpha \cdot \text{complexity}(T)$

Reference: Bertsimas, D., Dunn, J., Gibson, E., & Orfanoudaki, A. (2022). Optimal survival trees. *Machine Learning*, 111, 2951–3000.

Appendix

Likelihood Functions in Survival BSS

For each of the 2^p subsets, we fit the model via maximum likelihood

① Cox PH: $L_{\text{partial}} = \prod_{i:\delta_i=1} \frac{\exp(\beta^\top X_i)}{\sum_{j:T_j \geq T_i} \exp(\beta^\top X_j)}$

Uses partial likelihood $\rightarrow$ baseline hazard $h_0(t)$ is never estimated

No distributional assumption on $T \rightarrow$ flexible, but sacrifices some efficiency

② AFT: $L_{\text{full}} = \prod_{i=1}^n [f(T_i|X_i)]^{\delta_i} [S(T_i|X_i)]^{1-\delta_i}$

Observed events ($\delta_i = 1 \rightarrow$ contribute **density** f

Censored observations ($\delta_i = 0 \rightarrow$ contribute **survival function** S

Requires a distributional assumption (Weibull, Log-normal, etc.)

Model Selection Criteria: AIC & BIC

Evaluating the 2^p Models: Information Criteria

As model gets complicated $\rightarrow$ Higher Likelihood $\rightarrow$ Penalize model complexity

① $AIC = -2 \log L + 2k$ k : number of estimated parameters ($\neq p$)
Balances goodness-of-fit against model complexity; favors larger models

② $BIC = -2 \log L + k \log(n)$
Stronger penalty than AIC for large $n \rightarrow$ tends to select sparser models

$$BIC_{\text{survival}} = -2 \log L + k \log(d)$$

model of Cox (1972), subjects contribute different amounts of information to the likelihood, depending on whether or not they are censored. We have found that, in the penalty term for BIC, substituting d , the number of uncensored events, for n , the total number of individuals, results in an improved criterion without sacrificing the asymptotic properties shown by Kass and Wasserman (1995).

Reference: Volinsky, C. T., & Raftery, A. E. (2000). Bayesian information criterion for censored survival models. *Biometrics*, 56(1), 256-262.

Replace n with d = number of observed (uncensored) events

Using $d < n$ reduces the penalty → more conservative, stable subset extraction proportional to the true information content

BSS selects the model with the smallest AIC or BIC